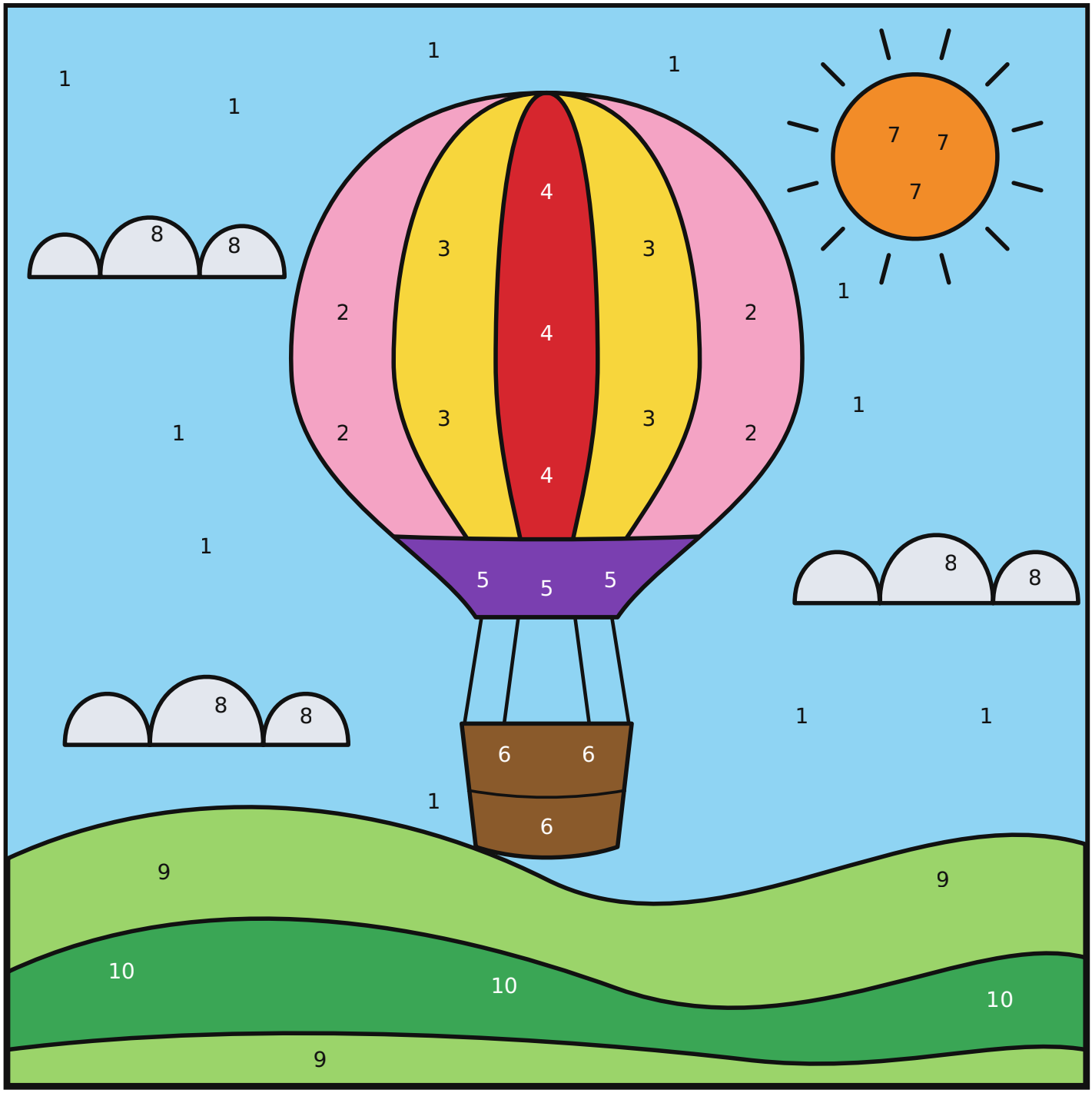
Name: **ANSWER KEY** _____ Date: _____

Linear Equations & Slope: Color By Number



Color Code:

2 Pink $y = -2x + 6$	**1** Light Blue $y = 2x + 3$	**3** Yellow $y = \frac{1}{2}x + 1$	**9** Light Green $y = -\frac{1}{2}x + 3$	**6** Brown $y = 2x - 3$
7 Orange $y = 3x - 1$	**5** Purple $y = -\frac{1}{2}x + 1$	**10** Green $y = 2$	**8** Gray $y = 2x + 4$	**4** Red $y = 3x + 5$

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Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Write the equation of the line through (0, 3) and (2, 7) in slope-intercept form.

slope = $\frac{4}{2} = 2$

y-intercept = 3

y = 2x + 3

1. Color: **Light Blue**

2.
Write the equation of the line with slope −2 that passes through (1, 4).

$y - 4 = -2(x - 1)$

$y - 4 = -2x + 2$

$y = -2x + 6$

2. Color: **Pink**

3.
Write the equation of the line through (−2, 0) and (2, 2).

slope = $\frac{2}{4} = \frac{1}{2}$

use (2, 2): $2 = \frac{1}{2}(2) + b$, so b = 1

$y = \frac{1}{2}x + 1$

3. Color: **Yellow**

4.
Write the equation of the line parallel to y = 3x − 4 that passes through (0, 5).

parallel lines have the same slope, 3

b = 5

y = 3x + 5

4. Color: **Red**

5.
Write the equation of the line perpendicular to y = 2x + 1 that passes through (4, −1).

perpendicular slope = $-\frac{1}{2}$

$-1 = -\frac{1}{2}(4) + b$, so b = 1

$y = -\frac{1}{2}x + 1$

5. Color: **Purple**

6.
Write 4x − 2y = 6 in slope-intercept form.

$-2y = -4x + 6$

$y = 2x - 3$

y = 2x − 3

6. Color: **Brown**

7.
A table of values has the points (0, −1), (1, 2) and (2, 5). Write the equation of the line.

slope = $\frac{3}{1} = 3$ (y goes up 3 each time)

y-intercept = −1

y = 3x − 1

7. Color: **Orange**

8.
A plant is 4 inches tall and grows 2 inches each week. Write an equation for its height *y* after *x* weeks.

start = 4, rate = 2 per week

y = 2x + 4

8. Color: **Gray**

9.
A line crosses the x-axis at (6, 0) and the y-axis at (0, 3). Write its equation.

slope = $\frac{3 - 0}{0 - 6} = -\frac{1}{2}$

$y = -\frac{1}{2}x + 3$

9. Color: **Light Green**

10.
Write the equation of the line through (−3, 2) and (5, 2).

slope = $\frac{0}{8} = 0$ (horizontal line)

y = 2

10. Color: **Green**